🌿 CLASS 10 SCIENCE: OUR ENVIRONMENT

ECO-NOTES

🌱 Subject: Biology Masterclass | 🍃 Hinglish Edition | 🌳 Topper Focus

1. What is an Ecosystem? 🦋

English: All interacting organisms in an area together with the non-living components of the environment form an ecosystem.

Hinglish Concept: Ek aisa system jisme saare living organisms (plants, animals) aur non-living things (air, water, soil) ek saath mil kar interact karte hain, use **Ecosystem** kehte hain.

> 💡 **Real-life Example:** Ek pond (talab), forest, aquarium, ya crop field ek ecosystem hai. Aquarium ek **Artificial Ecosystem** hai, jabki forest ek **Natural Ecosystem** hai! 🌿🐟

2. Components of an Ecosystem 🍄

Ecosystem ke do main parts hote hain:

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      ECOSYSTEM (Prakriti ka Sansar) / \ Biotic Components Abiotic Components (Living / Jeevit) (Non-living / Nirjeev) - Producers (Plants) - Temperature / Taapmaan - Consumers (Animals) - Rainfall / Varsha - Decomposers (Fungi) - Soil, Air, Light
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A. Biotic Components (Living things 🦁🌱)

- **Producers (Autotrophs):** Jo apna khana khud banate hain using sunlight.
  Example: Green plants, Algae.
- **Consumers (Heterotrophs):** Jo khane ke liye producers par depend hote hain. (Herbivores, Carnivores, Omnivores, Parasites).
- **Decomposers (Saprotrophs):** Dead plants/animals ko break down karte hain.
  Example: Fungi, Bacteria. (Nature ke scavengers 🦠).

Components of an Ecosystem (Contd.) 🌳

B. Abiotic Components (Non-living things ☀️💧)

Physical factors jaise **Temperature, Light, Wind, Soil, Water, aur Minerals**. Yeh decide karte hain ki wahan kis tarah ke flora (plants) aur fauna (animals) survive karenge.

3. Food Chain (Aahaar Shrinkhala) 🌾

Definition: A linear network of links in a food web starting from producer organisms and ending at apex predators.

Hinglish Concept: "Who eats whom" ki ek seedhi line ko food chain kehte hain jisme energy ek organism se dusre mein transfer hoti hai.

> 🌿 **Simple Food Chain Example:**
> Grass 🌾 (Producer) → Deer 🦌 (Herbivore) → Tiger 🐅 (Carnivore)

Key Characteristics:

- Energy flow hamesha **Unidirectional** (ek hi direction mein) hoti hai.
- Yeh 3 to 4 steps (trophic levels) se zyada lambi nahi hoti kyunki aage jaate-jaate energy khatam ho jati hai.

4. Food Web (Aahaar Jaal) 🕸️

Definition: An intricate network of many food chains intersecting with each other.

Hinglish Concept: Jab bohot saari food chains aapas mein jud kar ek complex jaal bana leti hain, toh use **Food Web** kehte hain. Yahan animals ke paas multiple choices hote hain.

5. Trophic Levels (Poshak Star) 📐

Food chain ke har step ya level ko **Trophic Level** kehte hain.

- **T_1 (First Trophic Level):** Producers (Plants 🌿)
- **T_2 (Second Trophic Level):** Herbivores (Primary Consumers 🦌)
- **T_3 (Third Trophic Level):** Carnivores (Secondary Consumers 🦊)
- **T_4 (Fourth Trophic Level):** Large Carnivores (Tertiary Consumers 🐅)

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T4: Large Carnivores (Tiger / Hawk) ▲ T3: Carnivores (Frog / Fox) ▲ T2: Herbivores (Deer / Rabbit) ▲ T1: Producers (Green Plants 🌿)
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6. 10% Law of Energy Flow (Lindeman's Law) ⚡

Definition: Only 10% of energy entering a trophic level is transferred to the next level. Baki 90% respiration aur heat ke roop mein lose ho jati hai.

> **Formula:** Energy at next level = 10% of previous level energy

10% Law: Solved Example & Trick 🎯

Agar Producer (T_1) ke paas **10,000 J** energy hai:

- **T_1 (Plants)** = 10,000 J
- **T_2 (Deer)** = 10% of 10,000 = 1,000 J
- **T_3 (Lion)** = 10% of 1,000 = 100 J
- **T_4** = 10 J

🚀 **Exam Shortcut Trick:** Har next level par jaane par ek zero (0) hata do!
10,000 → 1,000 → 100 → 10.

7. Biological Magnification (Biomagnification) 🧪

Definition: The progressive accumulation of harmful, non-biodegradable chemicals at each succeeding trophic level.

Hinglish Concept: Jaise-jaise hum food chain mein upar ki taraf jaate hain, harmful chemicals ki concentration barhti jati hai.

⚠️ **Exam Note: "Top = Toxic"**. Sabse zyada toxic chemicals Top Carnivores (Human/Lion) ke andar accumulate hote hain!

Biomagnification: Real-Life Application 🌍

Real-life Example:

- Fasal par DDT (pesticide) chhidka gaya 🌾💧
- Plants ne soil/water se absorb kiya (Low concentration 0.02 ppm)
- Herbivores ne khaya (Concentration badhi 0.5 ppm)
- Carnivores/Humans ne khaya (Maximum concentration/Toxic level 2.0+ ppm)

8. Ozone Layer Depletion ☀️🛡️

A. What is Ozone?

- **Formula:** O_3 (An isotope of oxygen).
- **Location:** Stratosphere (15-50 km above earth surface).
- **Role:** Sun ki harmful **Ultraviolet (UV) Radiations** ko absorb karti hai aur skin cancer, cataract se bachati hai.

Ozone Layer Depletion (Contd.) ❄️

B. Ozone Hole & Causes

Hinglish Concept: Ozone layer patli ho rahi hai jiska main reason hai **CFCs (Chlorofluorocarbons)**.

Source of CFCs: Refrigerators, ACs, aur aerosol sprays 🧊.

Mechanism: CFCs stratosphere mein pahunch kar chlorine atoms release karte hain jo ozone (O_3) ko tod kar oxygen (O_2) mein badal dete hain.

> 💡 **UNEP Agreement (1987):** Duniya bhar ke countries ne CFC production ko freeze karne ka pledge liya under **Montreal Protocol**! 📜

9. Waste Management & Disposal ♻️

Hum jo waste produce karte hain, use do main categories mein divide kiya gaya hai:

Feature	Biodegradable Waste 🍂	Non-Biodegradable Waste ☢️
Meaning	Broken down by biological microorganisms.	Cannot be broken down easily.
Time taken	Fast (days to weeks).	Extremely slow (thousands of years).
Impact	Eco-friendly if disposed properly.	Causes severe pollution & biomagnification.
Examples	Vegetable peels, paper, cow dung.	Plastics, polythene bags, glass, DDT.

Waste Management Strategies 🔄

Wastes ko manage karne ke liye **3 R's principle** follow kiya jata hai:

- **Reduce:** Kam se kam waste generate karna (jaise paani/bijli bachana, extra packaging avoid karna).
- **Reuse:** Cheezon ko dobara use karna bina unhe fenke (jaise plastic bottles ko plant pot banana 🪴).
- **Recycle:** Waste material ko process karke nayi useful cheezein banana (jaise old paper se recycled paper 📄).

⭐ **Topper Tip:** "Reduce" sabse best option hai kyunki isme energy aur resources ki sabse kam wastage hoti hai.

10. Diagrams & Flowcharts Explanation 📊

Text-Based Diagram: Energy Flow in an Ecosystem

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      Sun ☀️ | ▼ (Light Energy) [ Producers (Plants 🌿) ] ──(90% Heat Loss)──> | ▼ (10% Energy Passed) [ Herbivores (Deer 🦌) ] ──(90% Heat Loss)──> | ▼ (10% Energy Passed) [ Carnivores (Tiger 🐅) ] ──(90% Heat Loss)──>
```

Explanation: Energy ka flow **ek-tarfa (unidirectional)** hota hai. Sun se energy plants mein aati hai, aur phir step-by-step consumer levels par kam hoti jati hai. Energy kabhi wapas Sun par nahi jaati.

11. Exam Short Tricks & Mnemonics 💡

🚀 **Topper's Quick Tricks:**

- **Ozone Formula Trick:** O ke baad 3 aata hai → **O_3**.
- **10% Law Trick:** Har level par ek zero (0) hata do! (10,000 → 1,000 → 100 → 10).
- **Biomagnification Tip:** Yaad rakho **"Top = Toxic"**. Top trophic level par toxic chemical sabse zyada hoga.
- **Biodegradable vs Non-bio:** Bio = Living (jo decompose ho jaye). Non-bio = Man-made plastics (jo kabhi khatam na ho).

Frequently Asked Questions (Board Exam Focus) 🎯

- **Q1. Why are food chains generally limited to 3 or 4 trophic levels?**
  Ans: Because according to the 10% law, very little energy is available for the next level after 3-4 transfers, making survival impossible beyond that.
- **Q2. What would happen if we kill all organisms in one trophic level?**
  Ans: The preceding level population will grow uncontrollably, and the succeeding level will starve, collapsing the ecosystem.
- **Q3. Why is plastic non-biodegradable?**
  Ans: Because microorganisms lack the specific enzymes required to break down synthetic polymers found in plastics.

⚡ Quick Revision Checklist (Last-Minute Summary) ✅

- [ ] **Ecosystem:** Biotic (living) + Abiotic (non-living).
- [ ] **Producers:** Autotrophs (Plants 🌿).
- [ ] **Decomposers:** Cleaners of nature (Bacteria/Fungi 🦠).
- [ ] **10% Law:** Lindeman ne diya tha; sirf 10% energy next level par jati hai.
- [ ] **Biomagnification:** Chemicals ki concentration trophic level ke sath barhti hai.
- [ ] **Ozone (O_3):** UV rays se bachata hai; CFCs isko destroy karte hain.
- [ ] **Montreal Protocol:** Related to Ozone layer protection 📜.
- [ ] **3 R's:** Reduce, Reuse, Recycle ♻️.

🎉 All the Best for Your Board Exams! Score 100/100! 🌟